

Course Title	Digital Signal Processing Hardware		
Course Code	ED311	Credit Structure	3-0-2-4 (L-T-P-C)
Category	Core	Semester V	Winter Semester (AY 25-26)
Program	B.Tech (EVD)		
Prerequisites	Embedded Hardware Design		
Course Objectives/ Brief Course Description	Digital Signal Processing Hardware Digital system design and of digital signal processing in the context of modern systems. The course is intended to synthesise both sides (theory and experiment) of the module content, so as to broaden and deepen understanding. Syllabus • Introduction to DSP and DSP Hardware • Introduction to Stochastic Signals, Sampling and Quantisation • ADC, DAC Architectures • Registers, ALU, SRAM, IO ports, Instruction Decoder, Synthesis of a simplified MIPS microprocessor on FPGA • Testing combinational and sequential digital systems, Boundary Scan, Build-in self-test • Laboratory space and FPGA development kits required, Software requirements: Modelsim, Altera Quartus,		
Evaluation/ Grading Policy	2 In sem exams, 1 Final Exam, 1 Practical Session graded throughout the semester + Quizzes and Homeworks.		
Course Materials/ References	Textbooks J F Wakerly (2006). Digital Design - Principles and Practices. Pearson Prentice Hall. M M Mano, M D Ciletti, Digital Design (2007). Digital Design. Pearson Prentice Hall.		

Course Outcome:

Having successfully completed this module the student will be able to:

- Build and debug a digital circuit, FPGA and hardware implementations of combinational and sequential digital systems
- Signals, quantisation and coding of communications signals